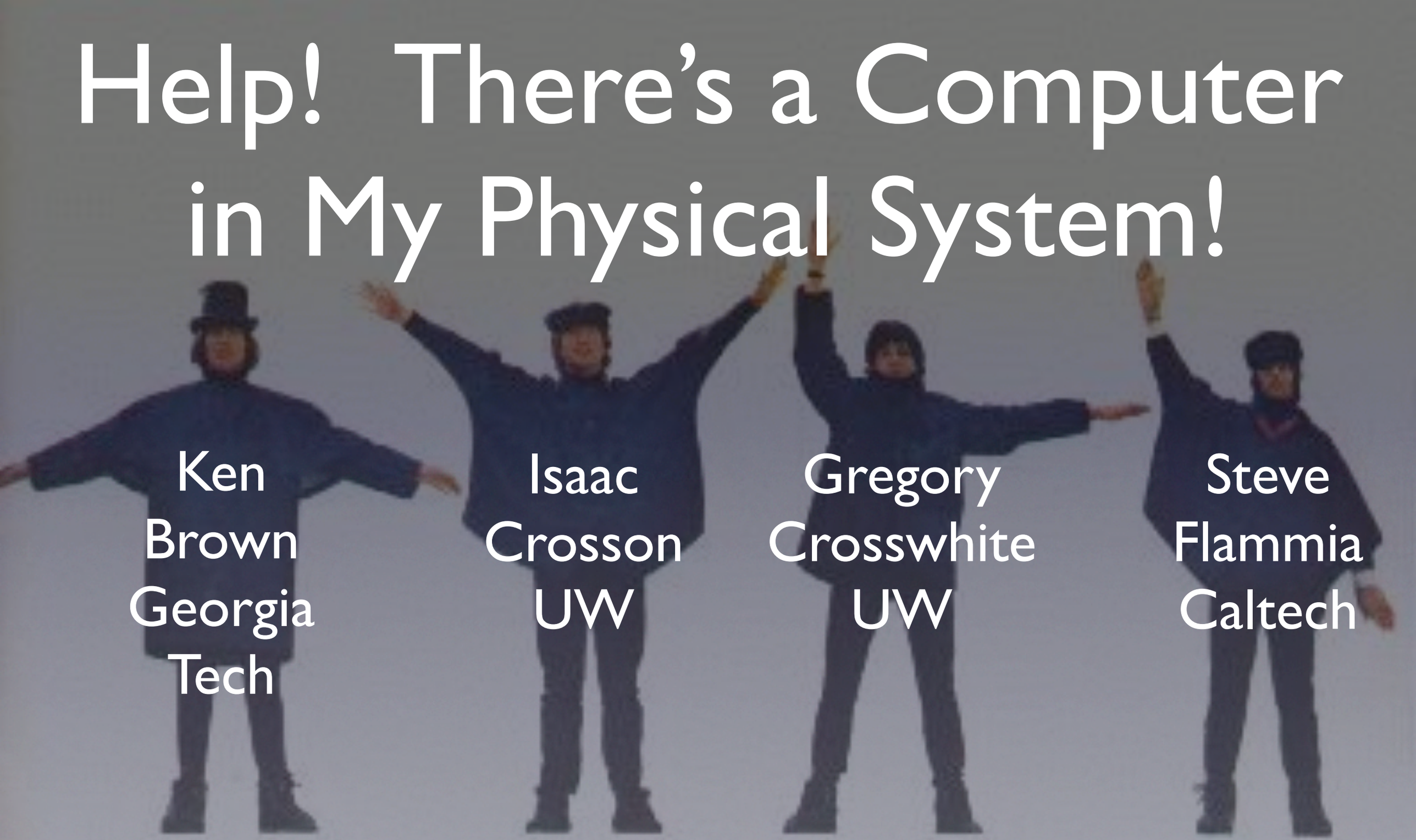
Help! There's a Computer in My Physical System!



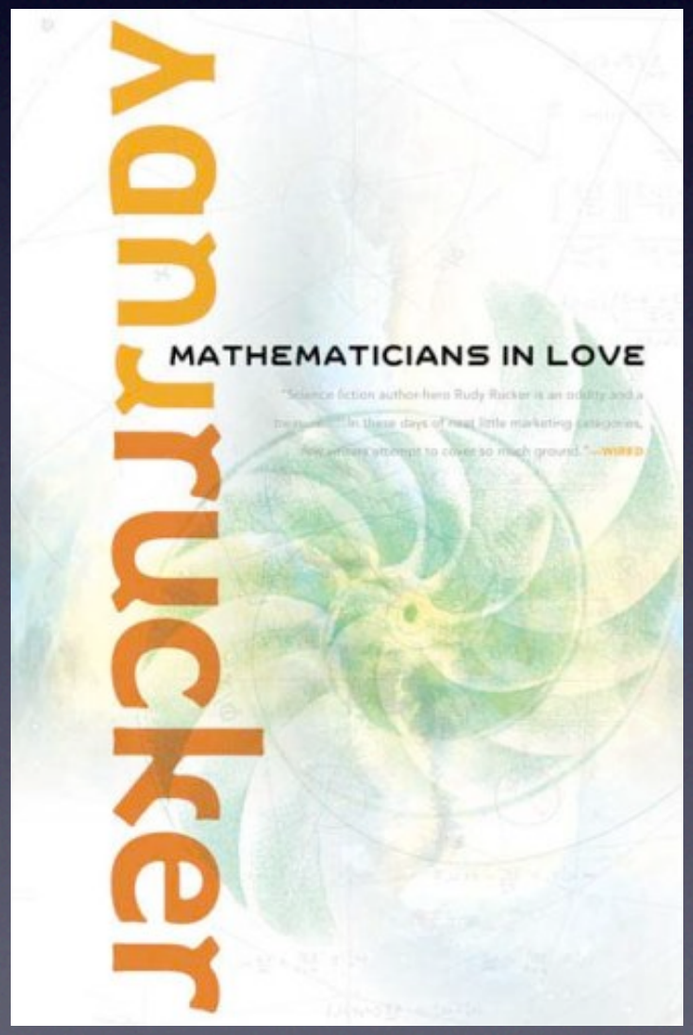
Dave Bacon
University of Washington



Warning!



The work described in this talk was inspired by the novel "Mathematicians in Love."

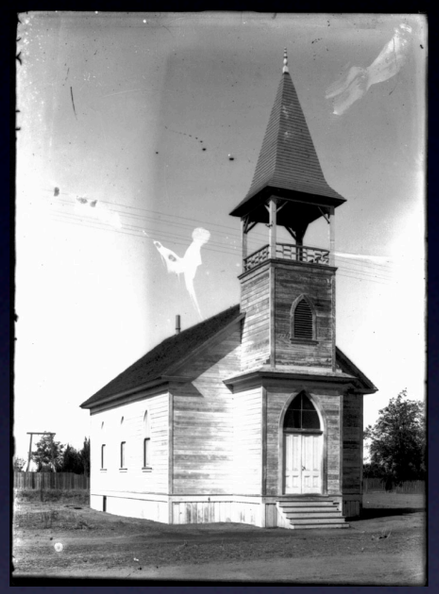


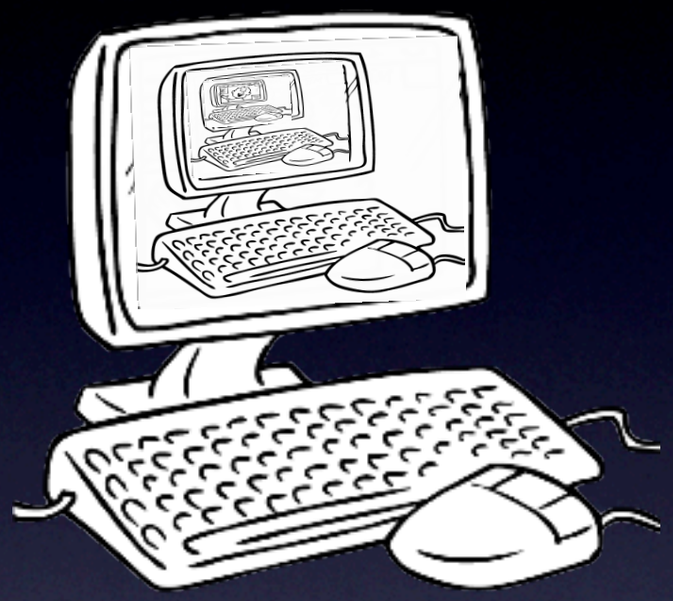


Rudy Rucker near a portal to another dimension.

Motivations

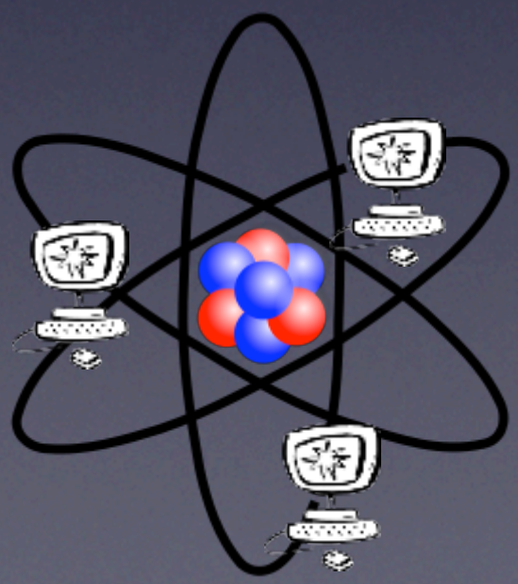
Computers can efficiently simulate other computers.







Physical systems that act like computers should be easy to simulate.



Cook-Levin Theorem



Theorem: the **Boolean Satisfiability Problem** is **NP-complete**

Boolean formula (CNF):

Boolean variables $\{x_1, \ldots, x_n\}$

$$\phi = C_1 \wedge C_2 \wedge \cdots \wedge C_k$$

$$C_i = (t_{i1} \vee t_{i2} \vee \cdots \vee t_{il_i})$$

$$t_{ij} \in \{x_1, \ldots, x_n, \bar{x}_1, \ldots, \bar{x}_n\}$$

Boolean Satisfiability: does there exist an assignment of $\{T, F\}$ to the boolean variables such that $\phi = T$?

Cook-Levin Theorem



Theorem: the **Boolean Satisfiability Problem** is **NP-complete**

NP: the class of languages such that $x \in L$ iff there is a certificate of membership that can be verified in polynomial time

$x \in L?$

certificate y



NP-complete: In NP and any problem in NP can be converted to the problem in polynomial time.

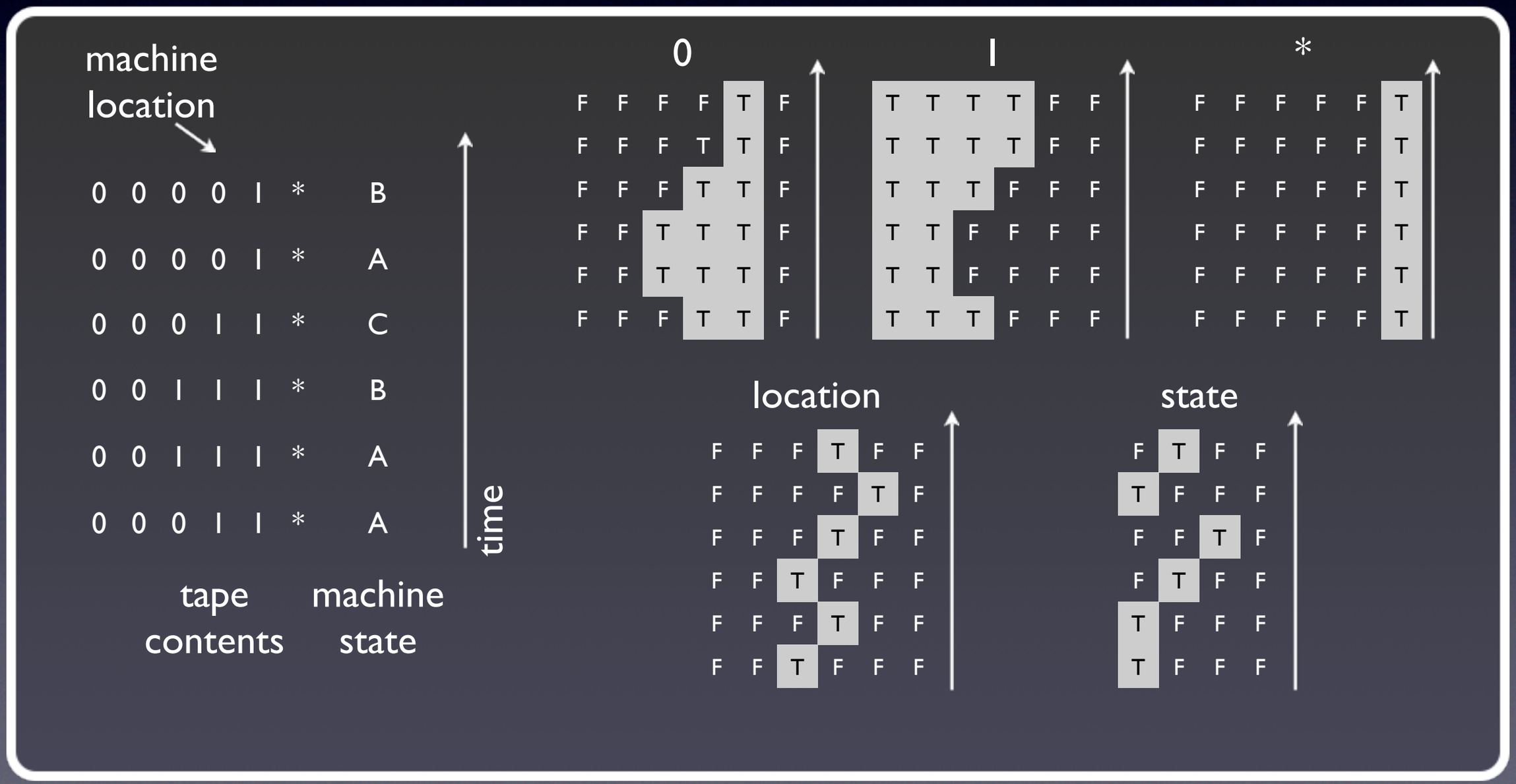
Cook-Levin Proof Sketch

NP-complete: In NP and any problem in NP can be converted to the problem in polynomial time.

Certificate is assignment of variables such that $\phi = T$ which can be used to verify in poly time

Cook-Levin Proof Sketch

NP-complete: In NP and any problem in NP can be converted to the problem in polynomial time.



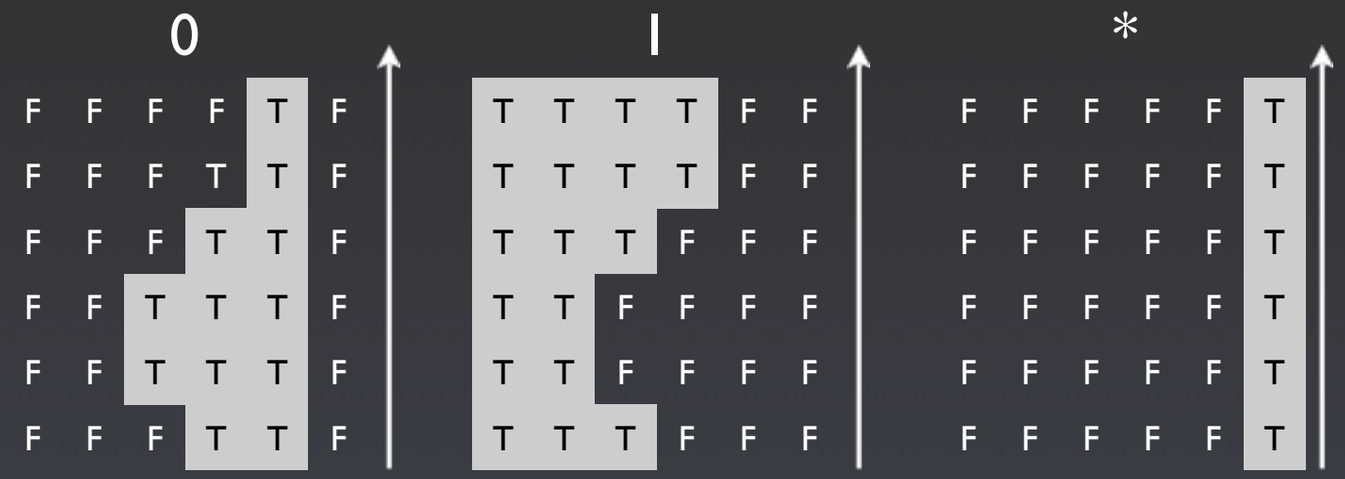
Cook-Levin Proof Sketch

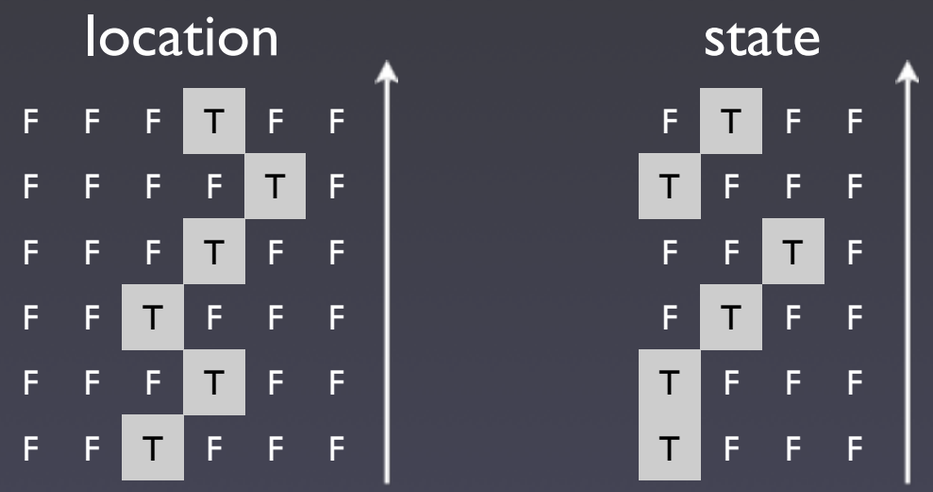
NP-complete: In NP and any problem in NP can be converted to the problem in polynomial time.

$T_{i,j,k}$ tape cell *i*, symbol *j*, step *k*

$H_{i,k}$ TM head at tape cell *i*, step *k*

$Q_{q,k}$ state *q*, step *k*





Cook-Levin Proof Sketch

NP-complete: In NP and any problem in NP can be converted to the problem in polynomial time.

$T_{i,j,k}$ tape cell *i*, symbol *j*, step *k*

$H_{i,k}$ TM head at tape cell *i*, step *k*

$Q_{q,k}$ state *q*, step *k*

$$T_{i,j,k} \rightarrow \neg T_{i,j',k} \quad (j \neq j')$$

one symbol per tape head

$$(H_{i,k} \wedge Q_{q,k} \wedge T_{i,\sigma,k}) \rightarrow$$
$$(H_{i+d,k+1} \wedge Q_{q',k+1} \wedge T_{i,\sigma',k+1}$$
$$(q', \sigma', d) = \delta(q, \sigma)$$

transition rule is obeyed

Cook-Levin Proof Sketch

NP-complete: In NP and any problem in NP can be converted to the problem in polynomial time.

$x \in L?$

certificate

y



$\Leftrightarrow$

$\bigwedge_i C_i$

True iff TM rules followed,
initial condition correct,
and machine accepts

Computer Science's "Wick Rotation" takes history of computation and turns it into an ahistorical SAT problem

**“Fixed on the front of her Fasbinder face
Was the kind of a smile
That only a rather dull child could have drawn
While attempting a graveyard in the moonlight...”**

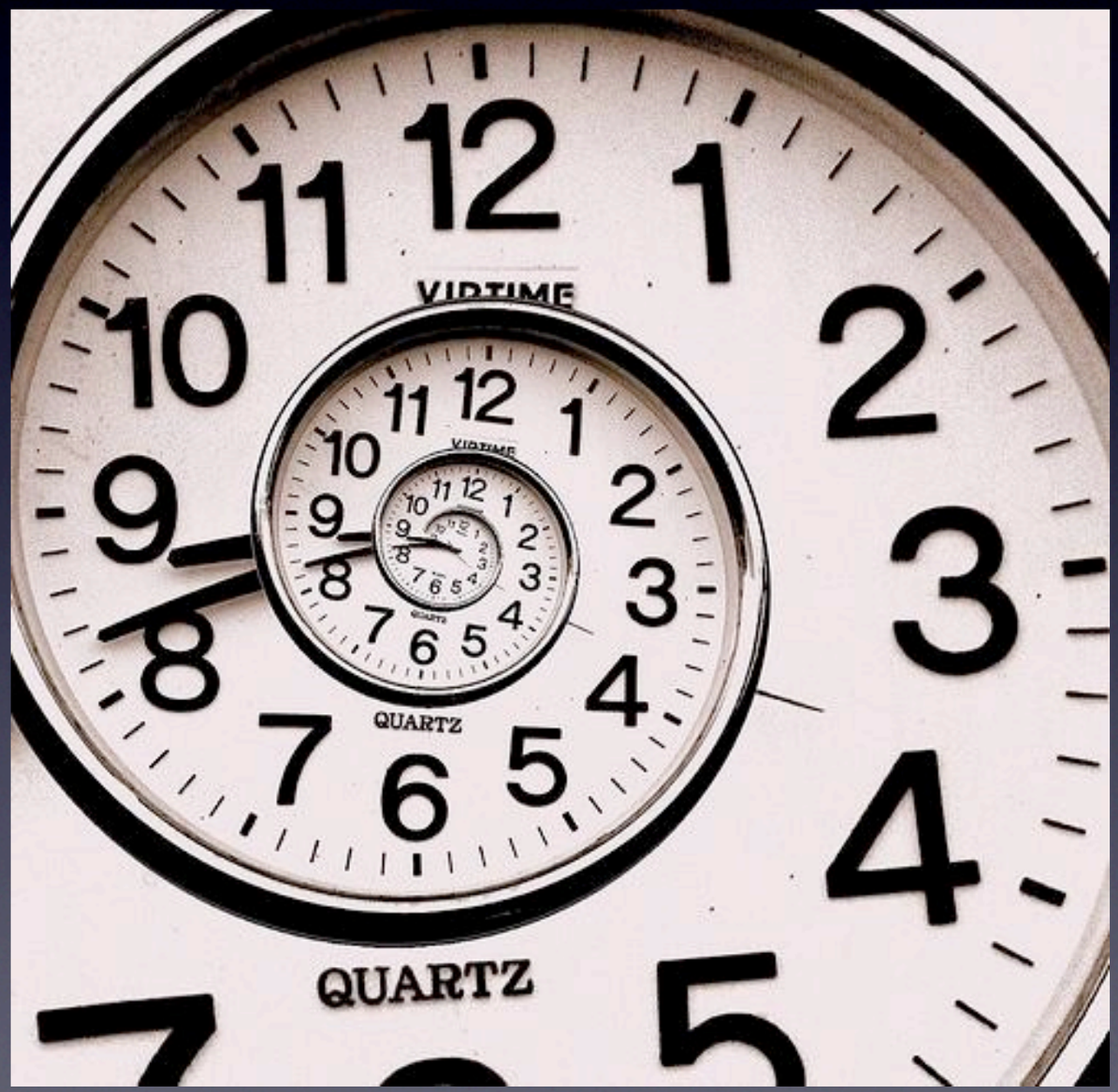
History Written Spatially

Physical systems in which “state” of the system is the history of a computation.

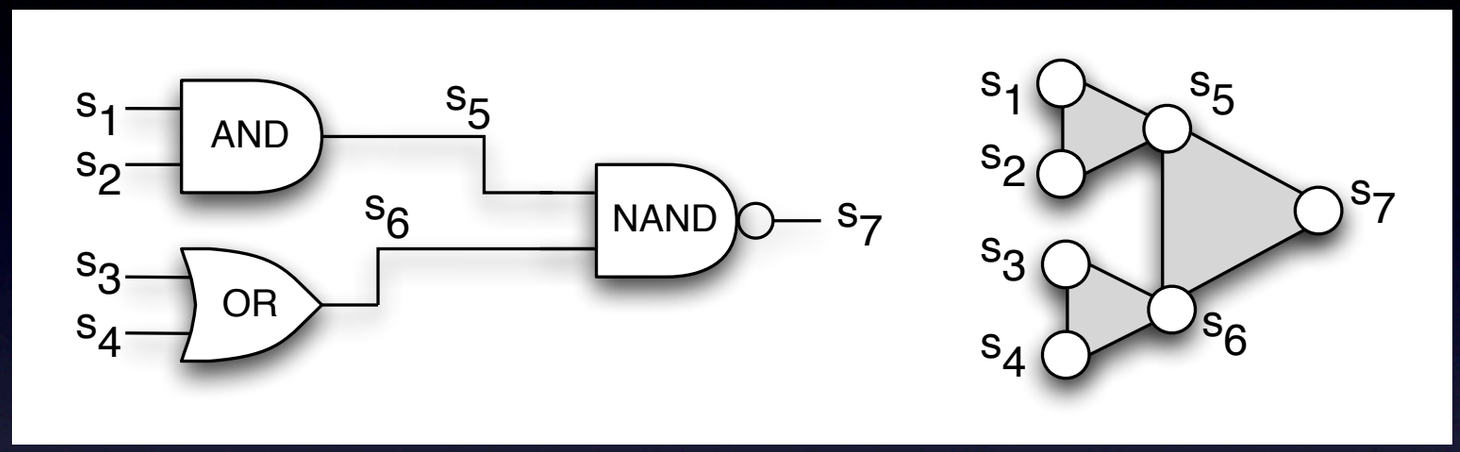
This Talk

- Finite temperature ground state classical computing
- Adiabatic cluster state quantum computing and adiabatic quantum transistors
- ....time permitting...
- Quantum density matrix renormalization group
- Graph state subsystem codes

Time Permitting = Please Talk To Me Afterwards!



Cook-Levin for Physicists



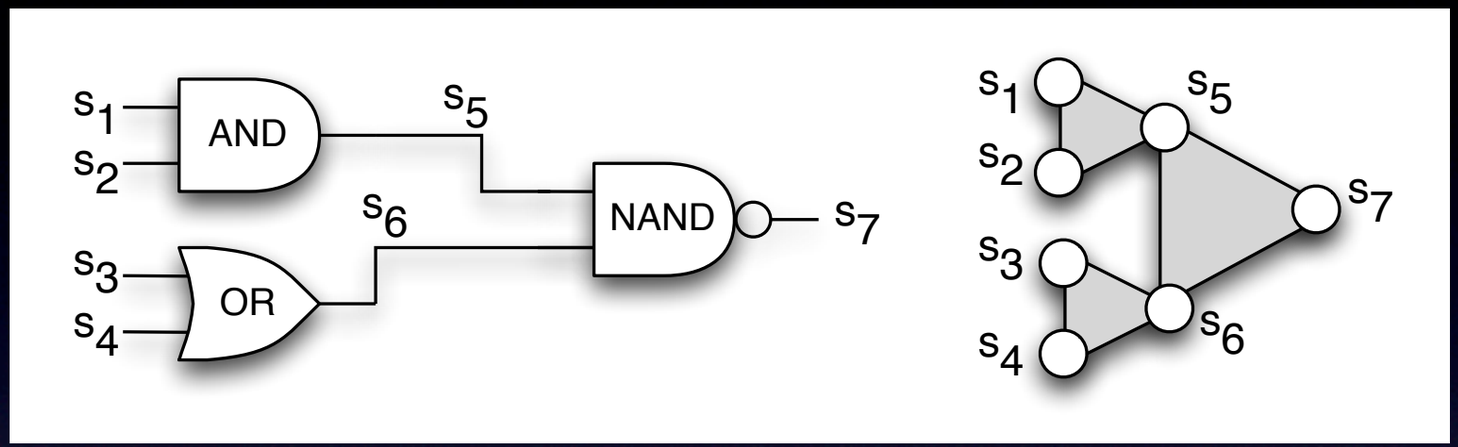
s_1	s_2	s_5	E_{AND}
0	0	0	0
0	0	1	Δ
0	1	0	0
0	1	1	Δ
1	0	0	0
1	0	1	Δ
1	1	0	Δ
1	1	1	0

$$E_{AND} = \Delta((1-s_1)(1-s_2)s_5 + (1-s_1)s_2s_5 + s_1(1-s_2)s_5 + s_1s_2(1-s_5))$$
$$= \Delta(s_5 + s_1s_2 - 2s_1s_2s_5).$$

$$E_C = \overbrace{\Delta(s_3 + s_4 + s_6 - s_3s_4 - 2s_3s_6 - 2s_4s_6 + 2s_3s_4s_6)}^{\text{OR}}$$
$$+ \underbrace{\Delta(s_5 + s_1s_2 - 2s_1s_2s_5)}_{\text{AND}} + \underbrace{\Delta(1 - s_7 - s_5s_6 + 2s_5s_6s_7)}_{\text{NAND}}$$

2^4-degenerate ground state = possible history of circuit

Cook-Levin for Physicists



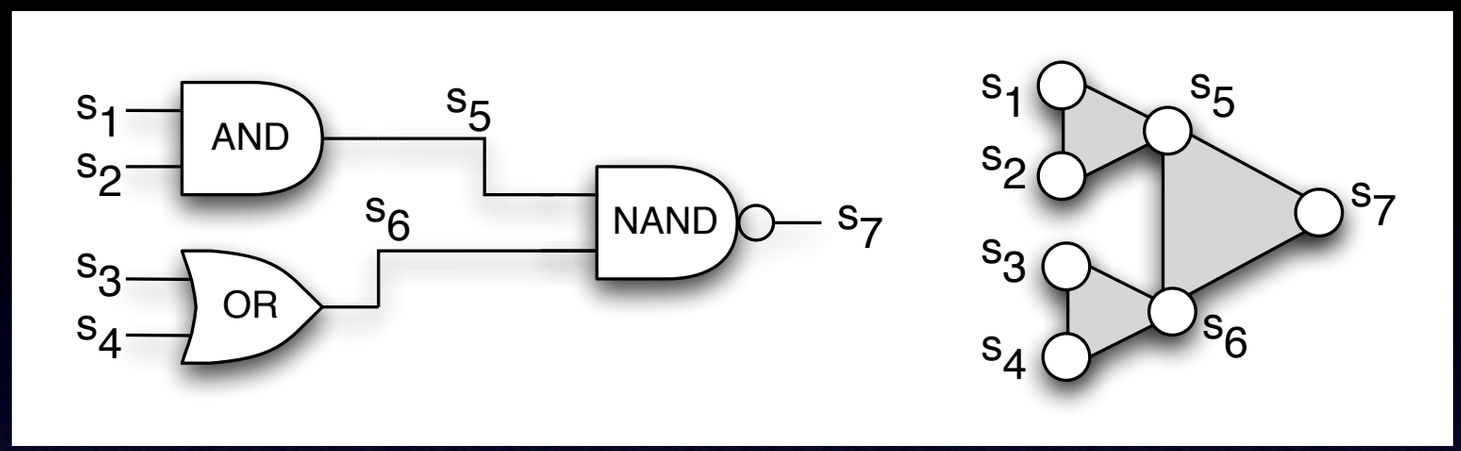
input: $s_1 = s_2 = s_3 = 0, s_4 = 1$

$$E_C = \overbrace{\Delta(s_3 + s_4 + s_6 - s_3 s_4 - 2 s_3 s_6 - 2 s_4 s_6 + 2 s_3 s_4 s_6)}^{\text{OR}} + \underbrace{\Delta(s_5 + s_1 s_2 - 2 s_1 s_2 s_5)}_{\text{AND}} + \underbrace{\Delta(1 - s_7 - s_5 s_6 + 2 s_5 s_6 s_7)}_{\text{NAND}}$$

$$E_{C,x} = E_C + \Delta(s_1 + s_2 + s_3 + (1 - s_4))$$

unique ground state is history of computation written out "spatially"

Ground State Computing



Energy = $\sum_{\text{gates}}$ (E=0 if g.s. executes gate, Δ otherwise)

+ $\sum_{\text{inputs}}$ (E=0 if input bit correct, Δ otherwise)

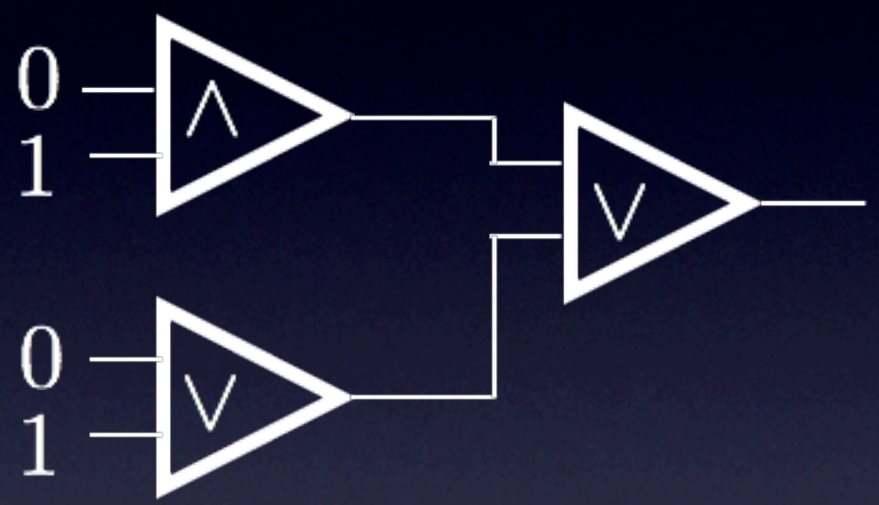
What happens if we consider these systems at finite temperature?

$$Pr(s_1, \ldots, s_n) = \frac{1}{Z} \exp(-\beta E(s_1, \ldots, s_n))$$

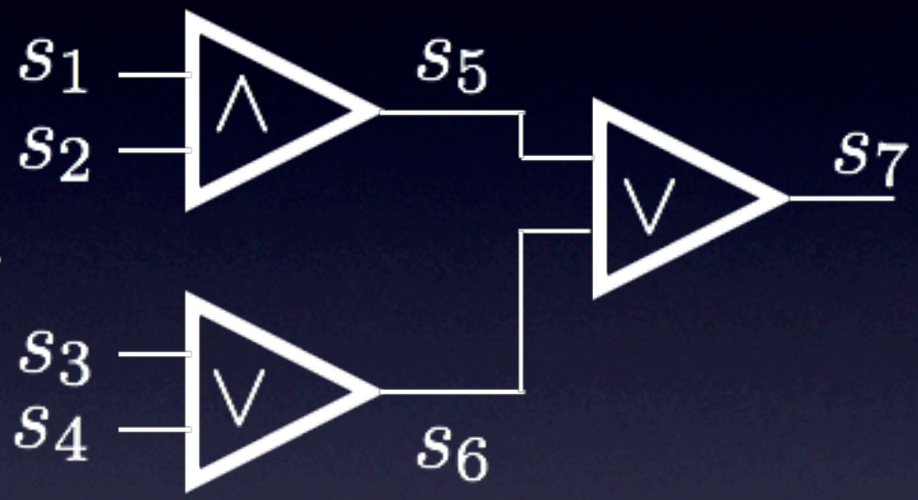
$$\beta = T^{-1}$$

Ground State Computing

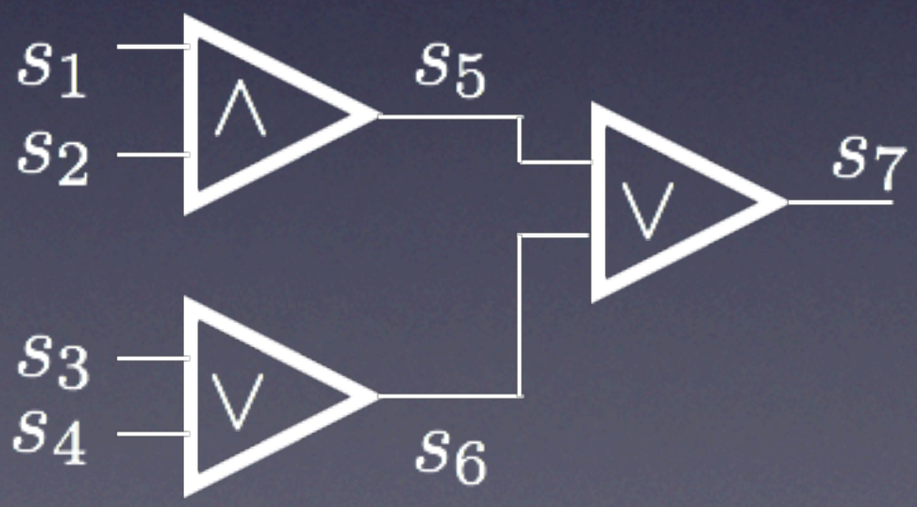
[Crosson, Bacon, and Brown, "Making Classical Ground State Spin Computing Fault-Tolerant", Phys. Rev. E, 82, 031106 (2010)]



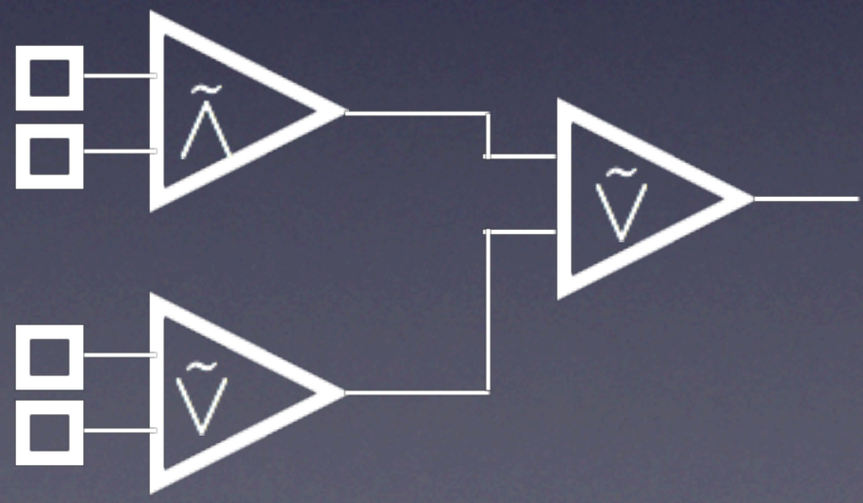
Deterministic circuit



Ground state computation

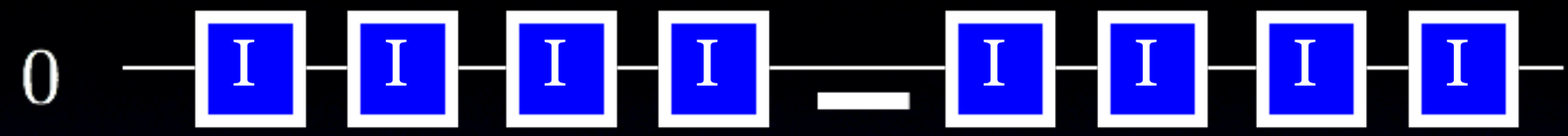


Thermal ground state computation



Probabilistic circuit

Fault-tolerant?



$$E_I(s_1,\ldots,s_{n+1}) = \Delta\delta_{s_1,1} + \Delta\sum_{i=1}^{n}\delta_{s_i,\neg s_{i+1}}$$

$$S_i = 1 - 2s_i$$

$$E_I(S_1,\ldots,S_{n+1}) = \frac{\Delta}{2}(1-S_1) + \frac{\Delta}{2}\sum_{i=1}^{n}(1-S_iS_{i+1})$$

One dimensional (ferromagnetic) Ising model with boundary term

Fault-tolerant?

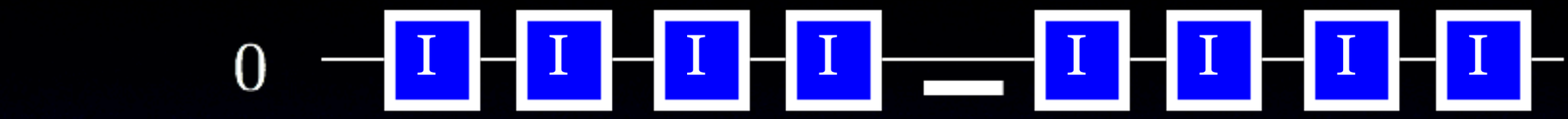


$$E_I(S_1,\ldots,S_{n+1}) = \frac{\Delta}{2}(1-S_1) + \frac{\Delta}{2}\sum_{i=1}^{n}(1-S_iS_{i+1})$$

$$M = \frac{1}{1+e^{-b}}\begin{bmatrix} 1 & e^{-b} \\ e^{-b} & 1 \end{bmatrix} \quad |v\rangle = \frac{1}{1+e^{-b}}\begin{bmatrix} 1 \\ e^{-b} \end{bmatrix} \quad b = \frac{\Delta}{T}$$

Define $|w\rangle = M^n|v\rangle$: the components of $|w\rangle$ are exactly the probabilities that the last bit is in the given state.

Fault-tolerant?



$$E_I(S_1,\ldots,S_{n+1}) = \frac{\Delta}{2}(1-S_1) + \frac{\Delta}{2}\sum_{i=1}^{n}(1-S_iS_{i+1})$$

$$M = \frac{1}{1+e^{-b}}\begin{bmatrix} 1 & e^{-b} \\ e^{-b} & 1 \end{bmatrix} \quad |v\rangle = \frac{1}{1+e^{-b}}\begin{bmatrix} 1 \\ e^{-b} \end{bmatrix} \quad b = \frac{\Delta}{T}$$

Probablistic gate

Probablistic preparation

$$|w\rangle = M^n|v\rangle$$

1. Not fault-tolerant.
2. Transfer matrix method = probabilistic classical circuit



Results

[Crosson, Bacon, and Brown, "Making Classical Ground State Spin Computing Fault-Tolerant", Phys. Rev. E, 82, 031106 (2010)]

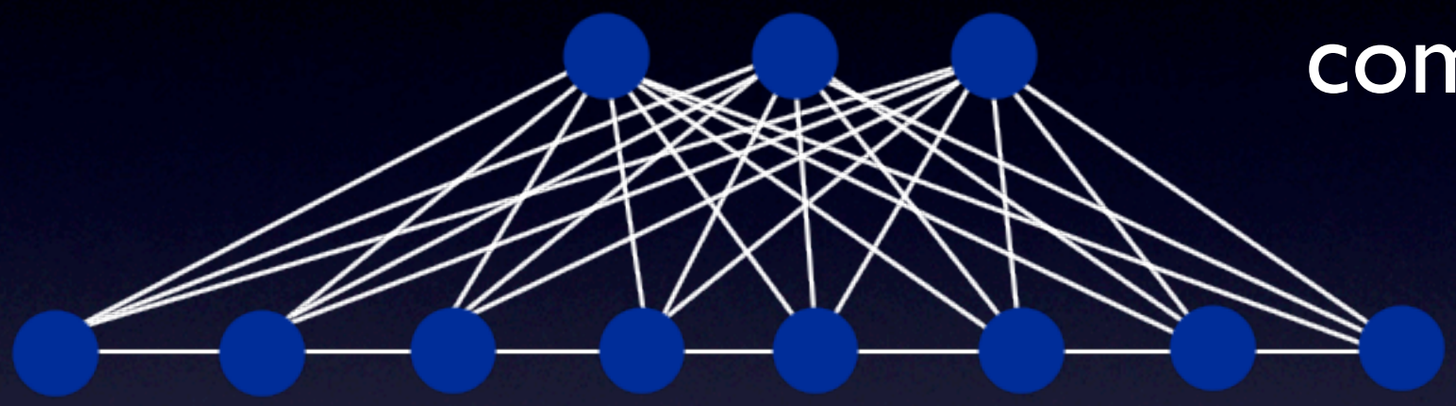
1. Not fault-tolerant in bare form
2. Ground state computing at T>0 is equivalent to classical gates being replaced by noisy versions
3. Use classical FT (von Neumman) to make FT models at T>0 (threshold = phase transition temperature)
4. NP-complete to determine whether generic spin model is a ground state computing model.
5. Promise-MA-complete problem:
let input bits float, constrain output with field
determine whether output bit is aligned with applied field with high probability

Dave, look at your watch

Quantum Ground State Computing

[Mizel, Mitchell, Cohen 2002, Mizel 2004]

[Aharonov *et al* 2004]



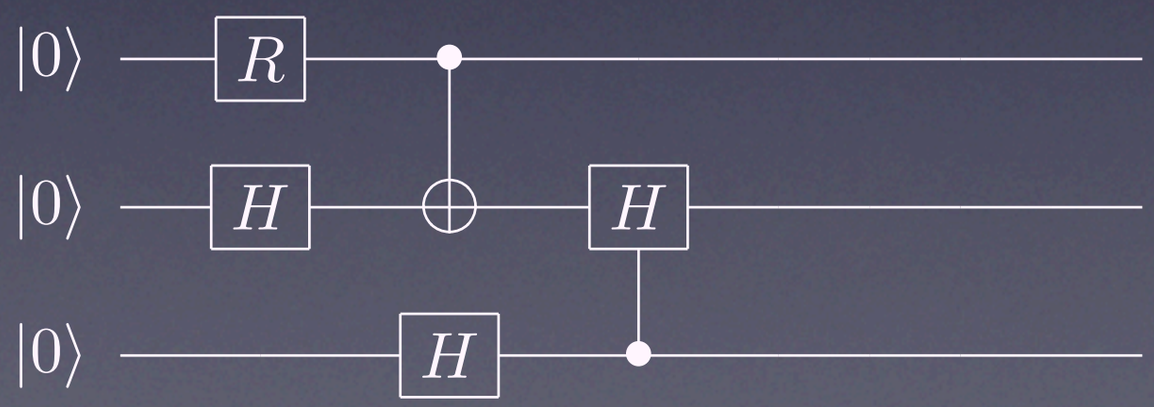
computational qubits

clock qubits

Adiabatically interpolate H_i to H_f

$$H(t) = \left(1 - \frac{t}{T}\right) H_i + \frac{t}{T} H_f$$

Prepare system into ground state



T polynomial in depth of circuit

Final ground state:

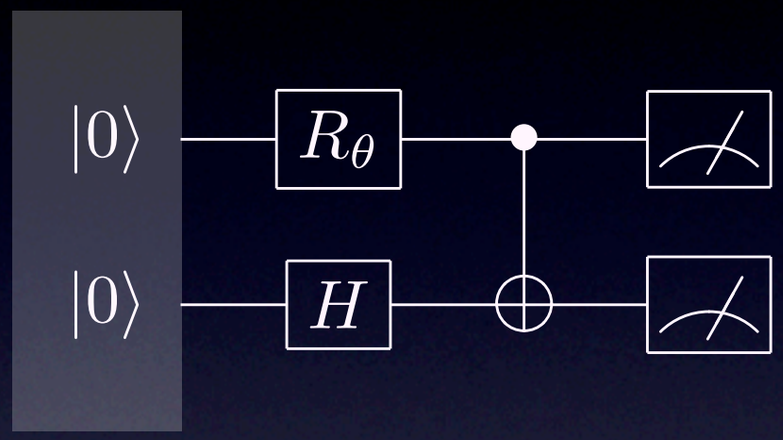
$$\frac{1}{\sqrt{T}} \sum_{t=1}^{T} |t\rangle_c V_t |0\rangle$$

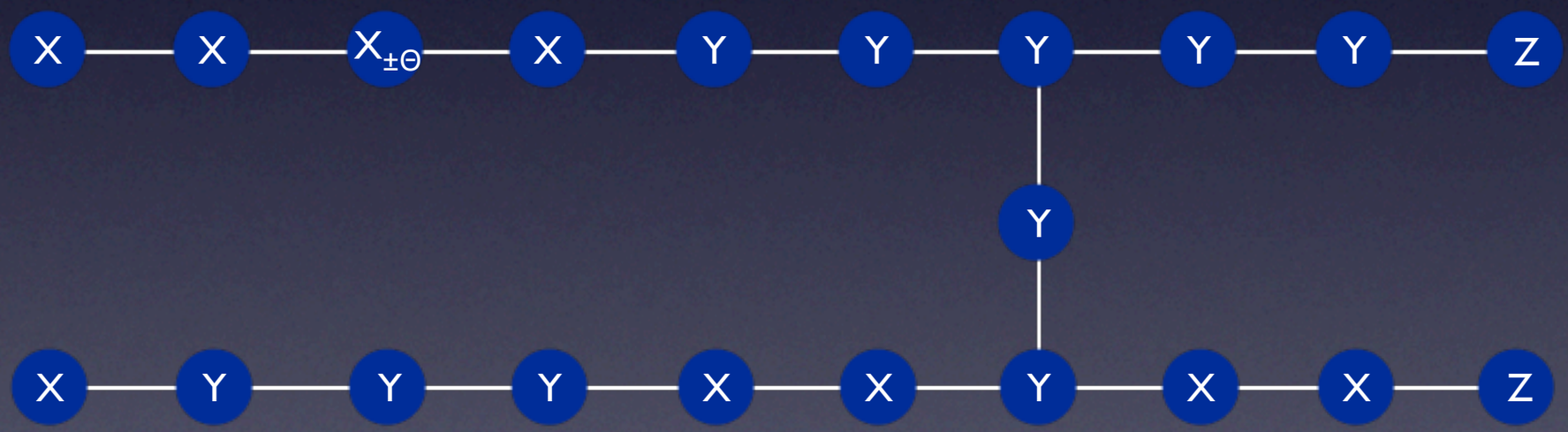
$$V_t = \prod_{i=t}^{1} U_i$$

One Way QC



[Raussendorf and Briegel, Phys. Rev. Lett. 86, 5188 (2001)]





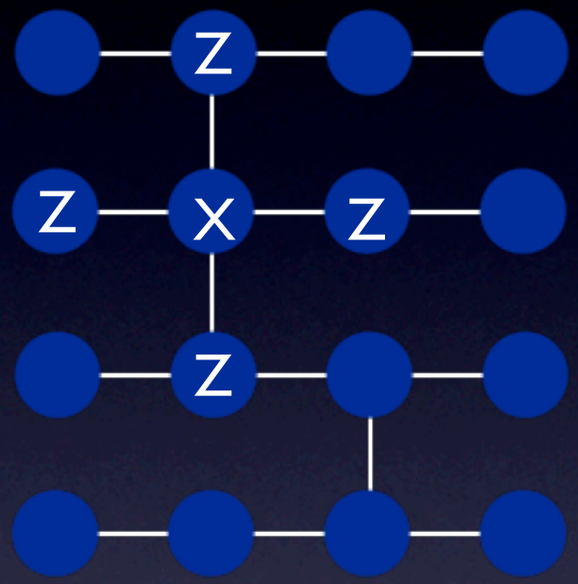
Adaptive measurement

Cluster state

Entangled state

circuit

Cluster State Hamiltonian



$$S_v = X_v \prod_{w \text{ adjacent to } v} Z_w$$

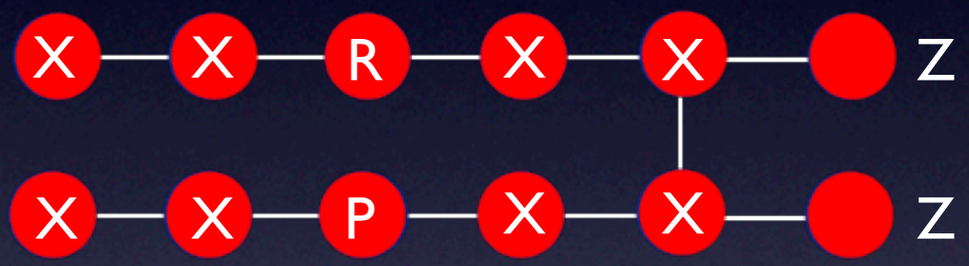
Cluster state is ground state of $H_C = -\Delta \sum_v S_v$

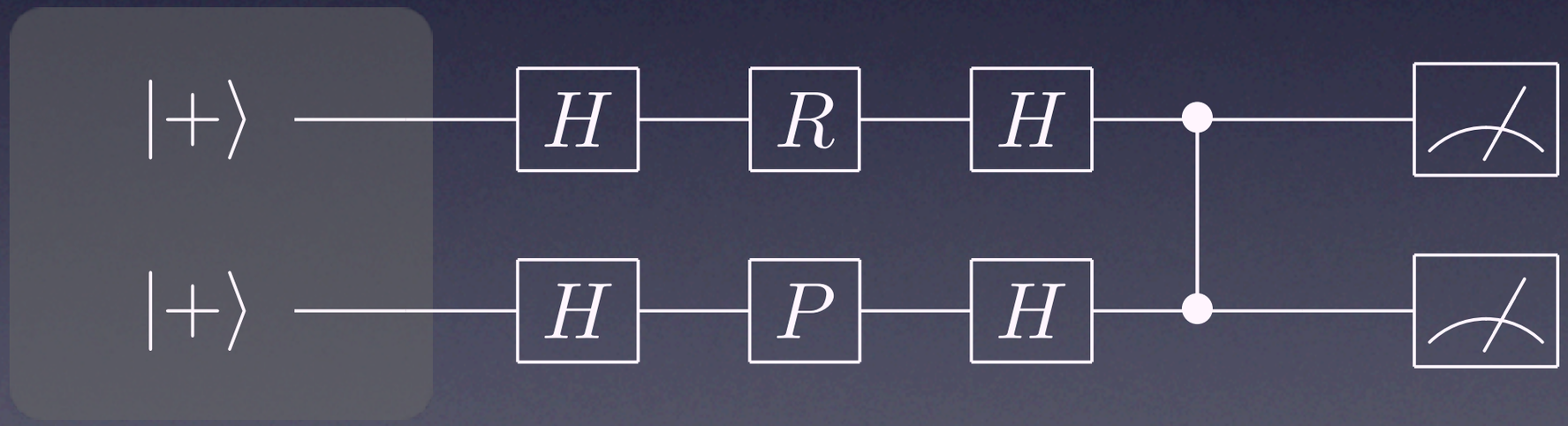
Possible to use gadgets to make only 2-qubit interactions:

[S. D. Bartlett and T. Rudolph, Phys. Rev. A 74, 040302 (2006)]

Adiabatic Cluster State QC

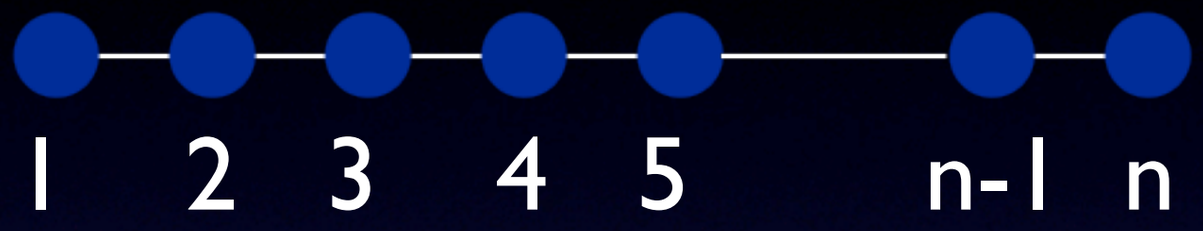
By turning on single qubit Hamiltonians while turning off parts of the cluster state Hamiltonian, we can enact a quantum circuit:





[Bacon and Flammia, "Adiabatic Cluster State Computing," arXiv:0912.2098 (2009)]

One Qubit



Modified cluster Hamiltonian:

$$H_C = -\Delta \sum_{i=1}^{n-1} S_i \qquad S_i = [Z]_i[X]_{i+1}[Z]_{i+2} \quad 1 \leq i \leq n-2$$

$$S_{n-1} = [Z]_{n-1}[X]_n$$

Ground state is 2-fold degenerate, can define logical operators for this qubit:

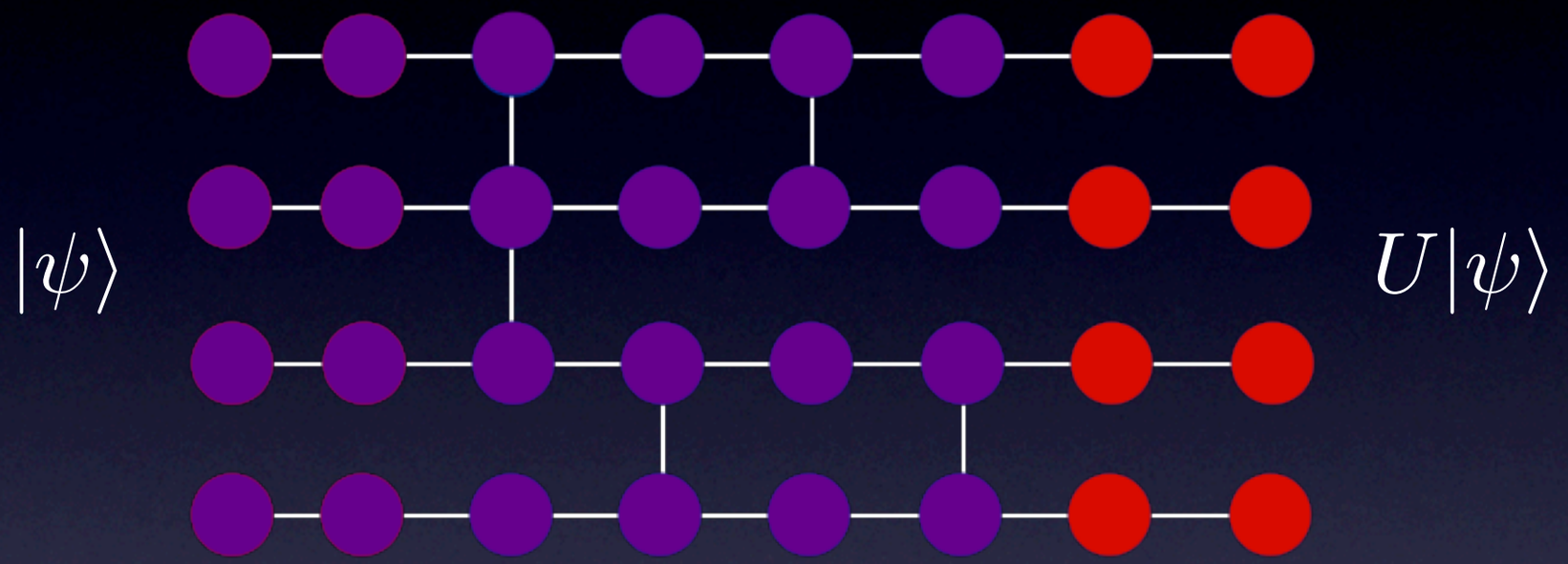
$$S_1 \rightarrow [X]_1$$

$$\bar{X} = [X]_1[Z]_2$$
$$\bar{Z} = [Z]_1$$

$$\bar{X} \cong [Z]_2$$
$$\bar{Z} \cong [X]_2[Z]_3$$

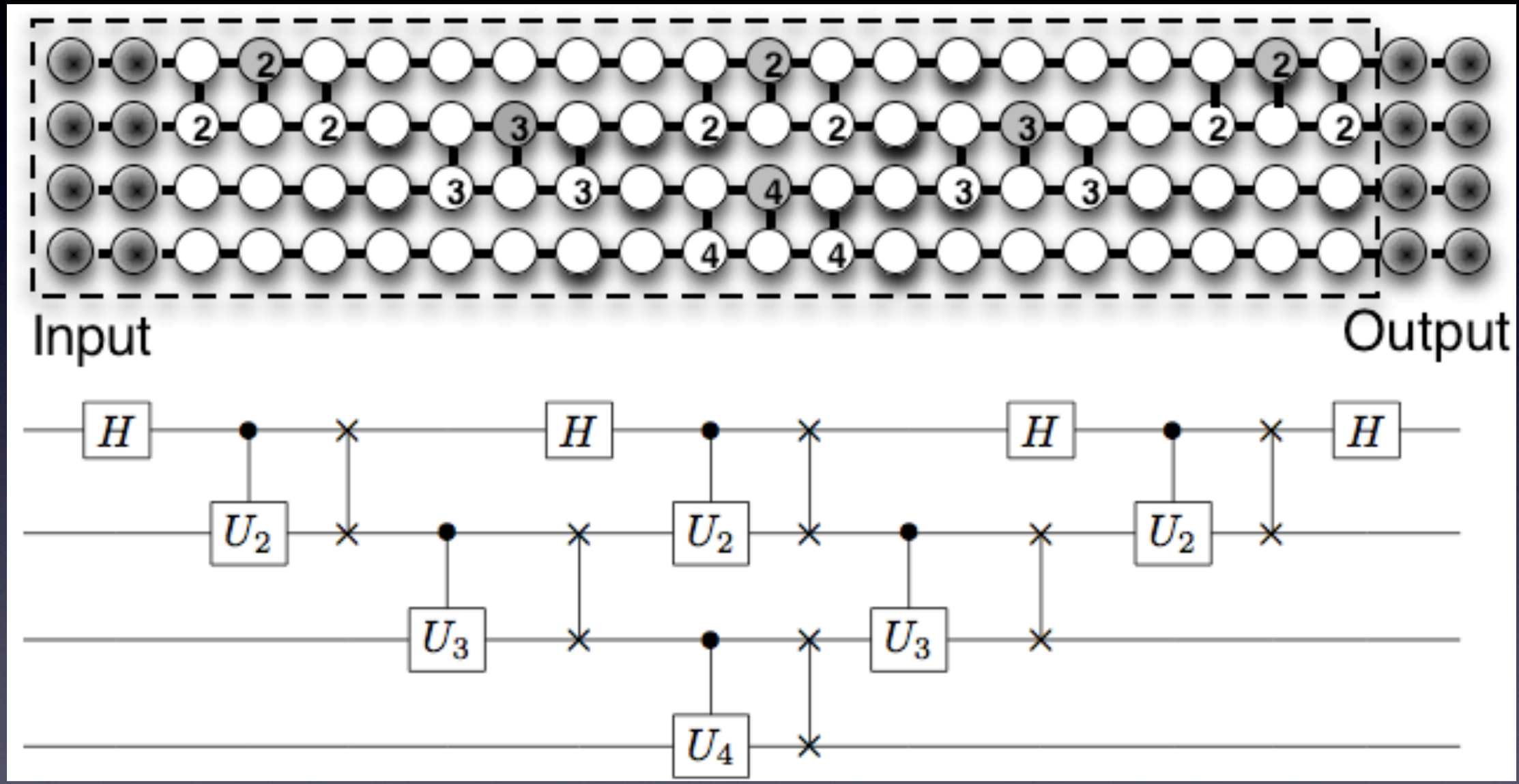
Quantum Transistor

[Bacon, Crosswhite, Flammia "Adiabatic Quantum Transistors" (2010) ask me for a copy]



1. Many-body system in its ground state
2. Qubits localized on one side of the device
3. Apply a strong 1-qubit external field to device
4. Qubits now localized on other side of device with a quantum circuit applied to the qubits

Example

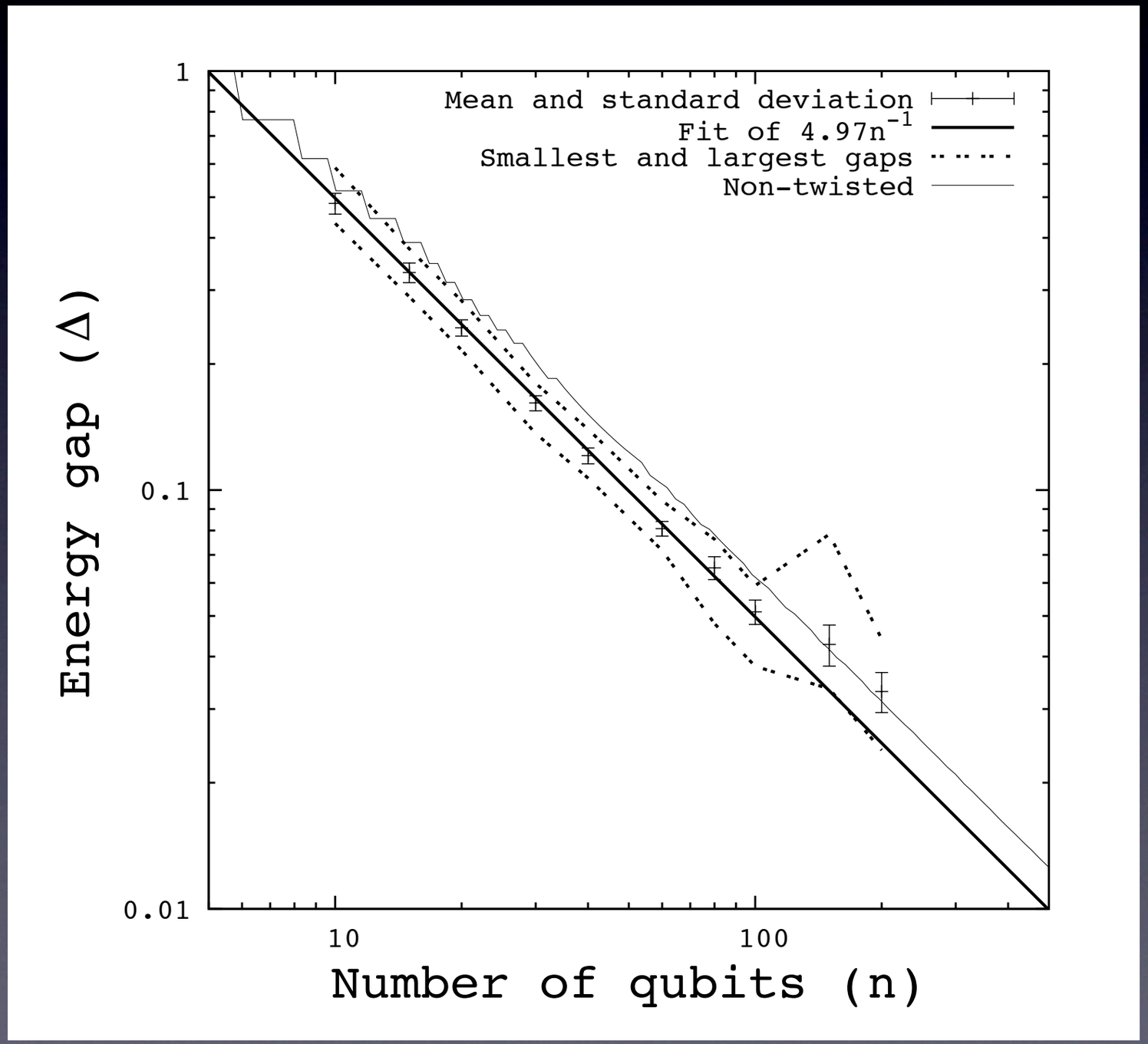


Adiabatic

- How slowly must we turn on the external field in order for the device to successfully quantum compute?

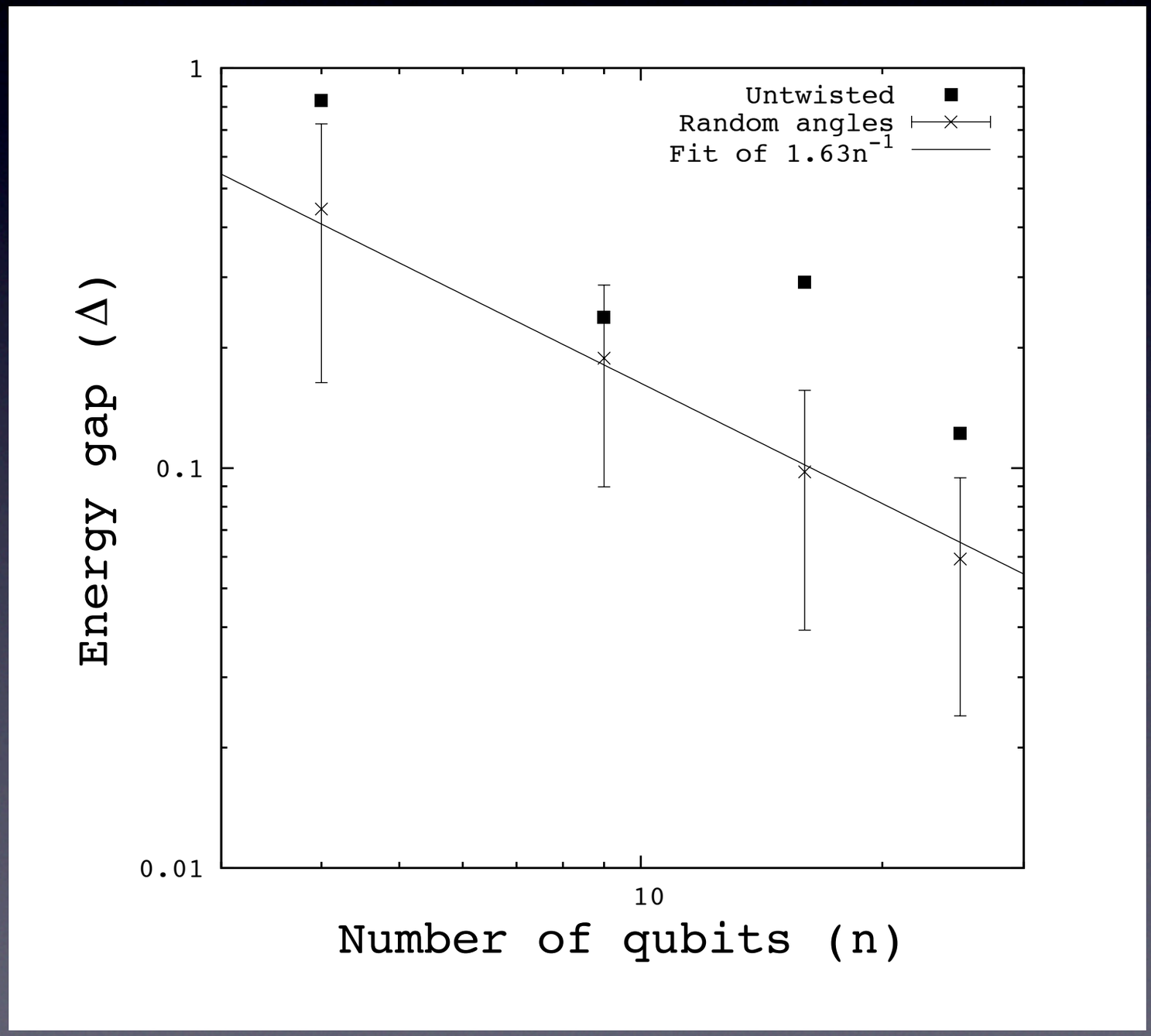
Adiabatic Quantum Transistor

Gap for 1D circuit with random angle rotations...



Adiabatic Quantum Transistor

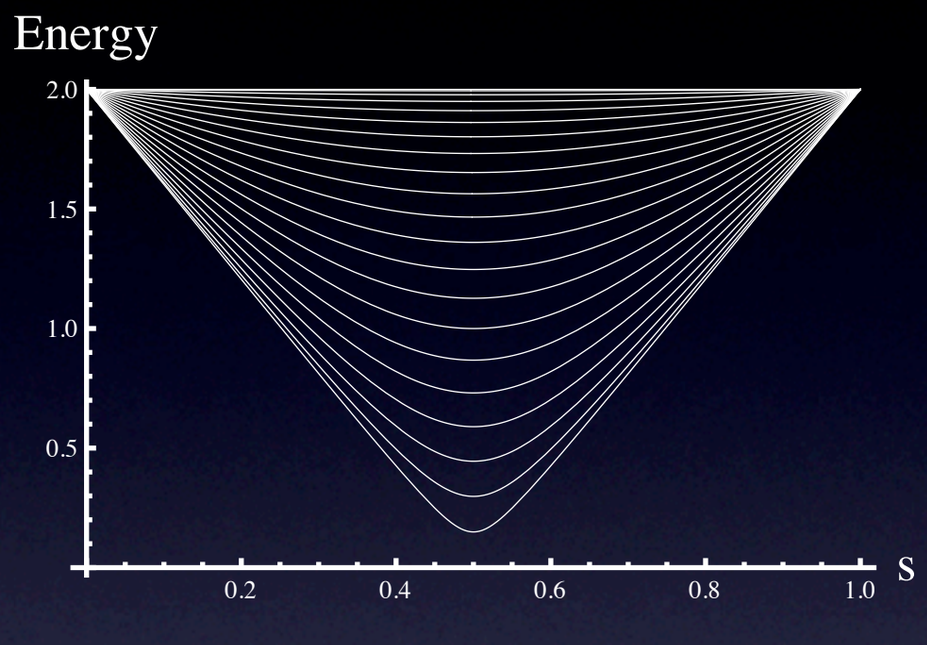
Gap for 2D circuit with random angle rotations...



Adiabatic

- How slowly must we turn on the external field in order for the device to successfully quantum compute?
  - 1D no twists, proven polynomial in size of circuit
  - 1D twists, extremely strong evidence polynomial in size of circuit
  - 2D twists, evidence of polynomial in size of circuit

Fault-Tolerance?



For 1D untwisted Hamiltonian, decouples into 2 1D Ising chains with transverse field.

Two types of errors:

- Those that change energy
- Those that act on degeneracy

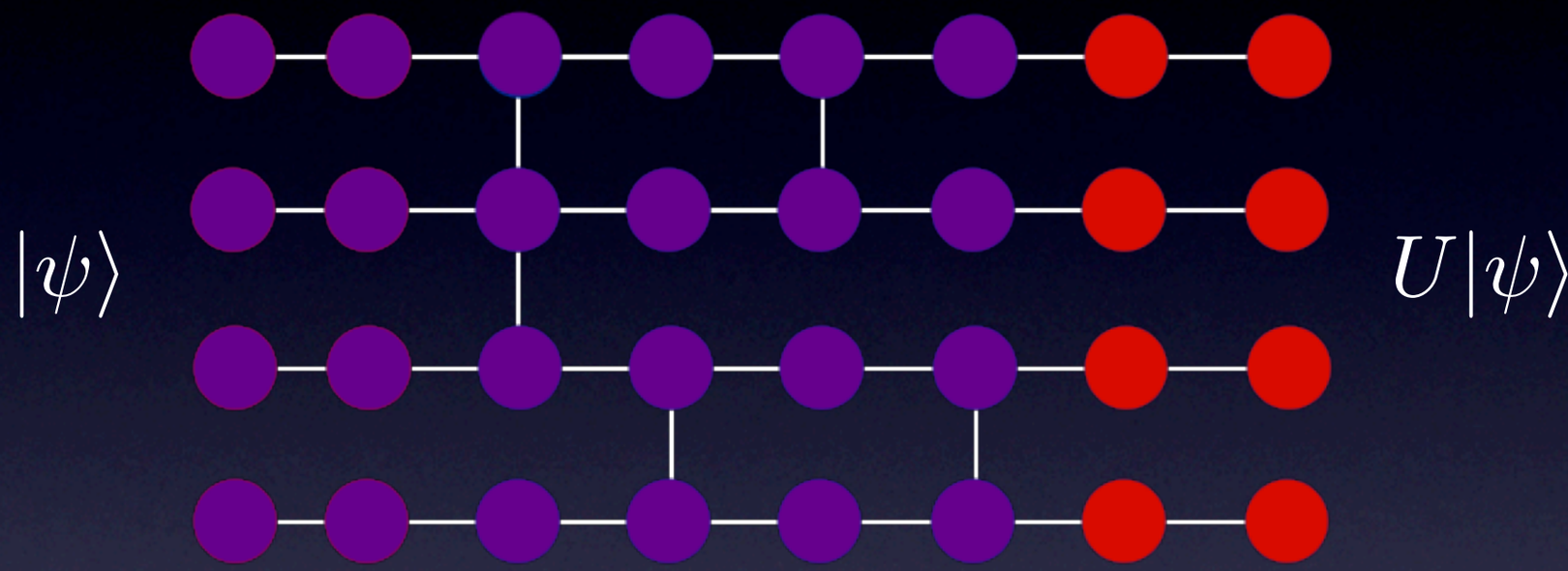
Which lead to

- Errors that change energy can be mapped to circuit model errors (independent error model)
- Degeneracy errors only don't cost energy at small and large times. Adjust schedule to make independent

1D errors mapped to independent error model

Quantum Transistor

[Bacon, Crosswhite, Flammia "Adiabatic Quantum Transistors" (2010) ask me for a copy]

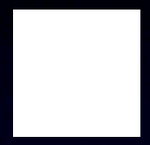


Computation is "encoded" spatially into Hamiltonian of system.
It is "enacted" by application of an applied field.

In middle of computation system is in superposition over computation.

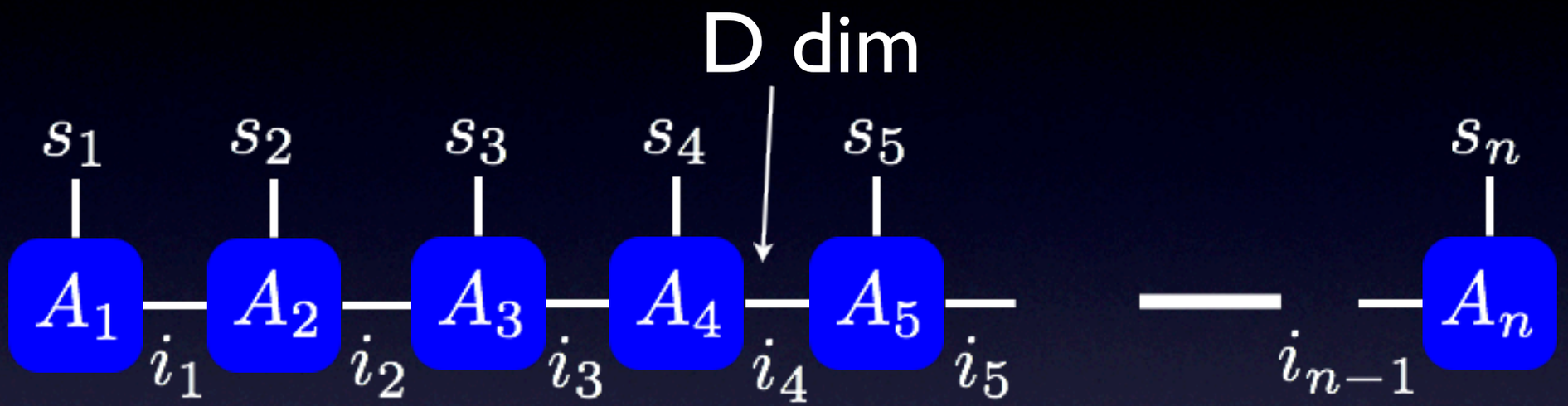
Quantum Transistor

"Clocked" architectures



Quantum Circuits in States?

Matrix product states (MPS):



$$\sum_{s_1,\ldots,s_n\in\{0,1\}} \sum_{i_1,\ldots,i_n\in\{1,\ldots,D\}} [A_1]^{s_1}_{i_1}[A_2]^{s_2}_{i_1,i_2}[A_3]^{s_3}_{i_2,i_3}\cdots[A_n]^{s_n}_{i_{n-1}}|s_1,\ldots,s_n\rangle$$

One can always impose

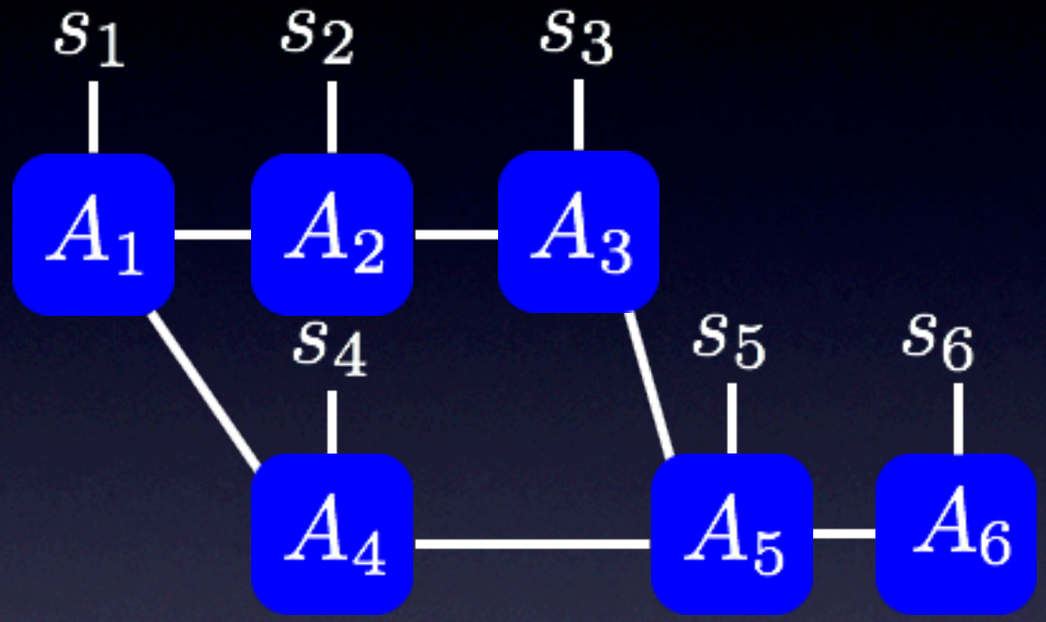
$$\sum_{s_k}[A_k^\dagger]^{s_k}[A_k]^{s_k} = I$$

i.e. we can interpret MPS as a quantum circuit in "correlation space"

[Gross, Eisert, Schuch, Perez-Garcia, arXiv:0706.3401]

Quantum Circuits in States?

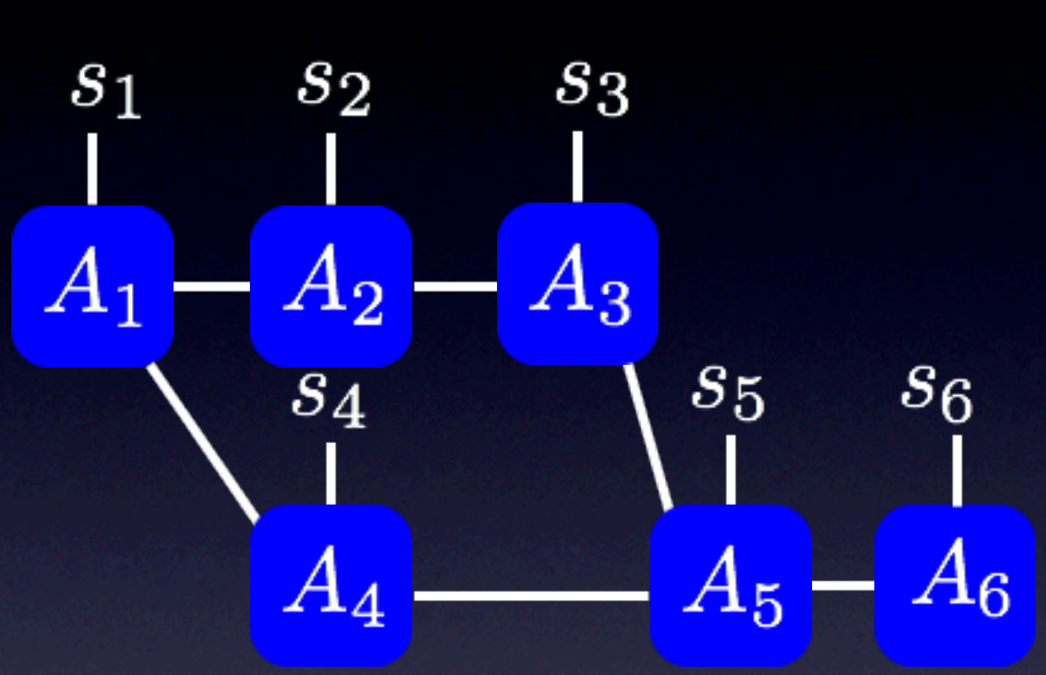
Tensor network states:

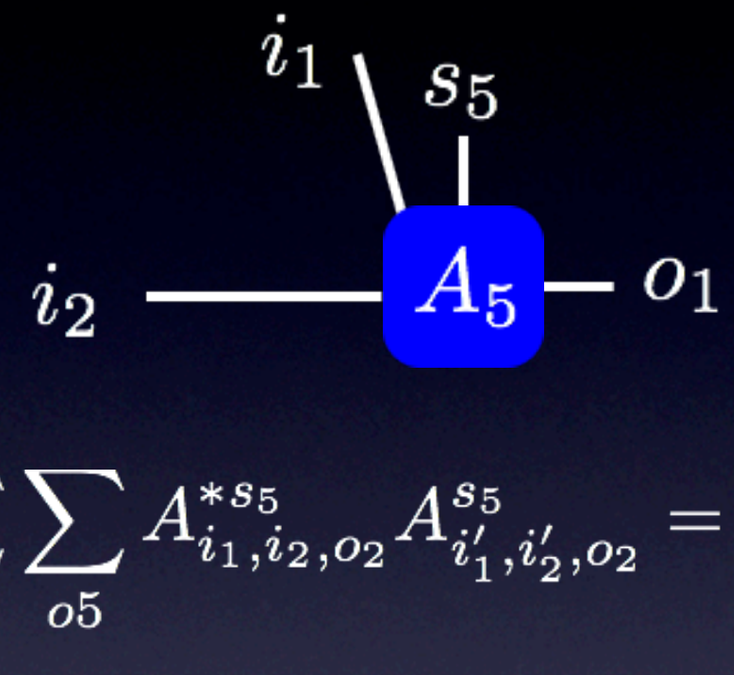


- Efficient representation
- Hard to compute expection values of local observables

Some classes of tensor network states allow efficient classical local observable calculations (MERA, etc)

Quantum Circuit Tensor States





$$\sum_{s_5} \sum_{o5} A^{*s_5}_{i_1,i_2,o_2} A^{s_5}_{i'_1,i'_2,o_2} = \delta_{i_1,i'_1} \delta_{i_2,i'_2}$$

A class of tensor states for which there exists an efficient quantum algorithm for calculating expectation values of local observables

Quantum Density Matrix Renormalization Group

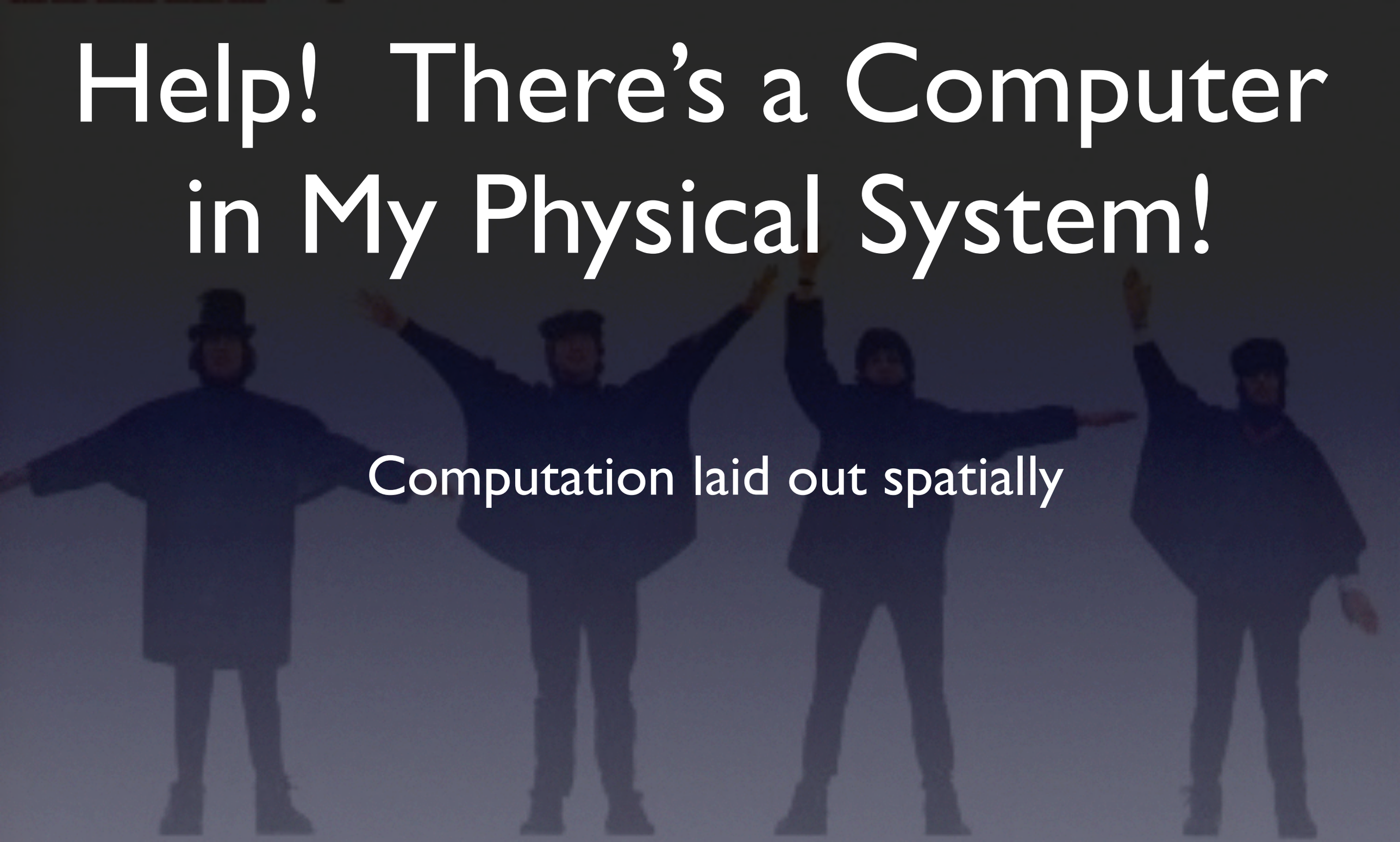
[D. Bacon and G. Crosswhite]

Heuristic (variational) quantum algorithm for finding ground states of many-body quantum systems.

Works well when ground state is well approximated by a quantum circuit tensor network state.

Quantum circuit tensor network states are complete, but are they really useful? Work in progress!

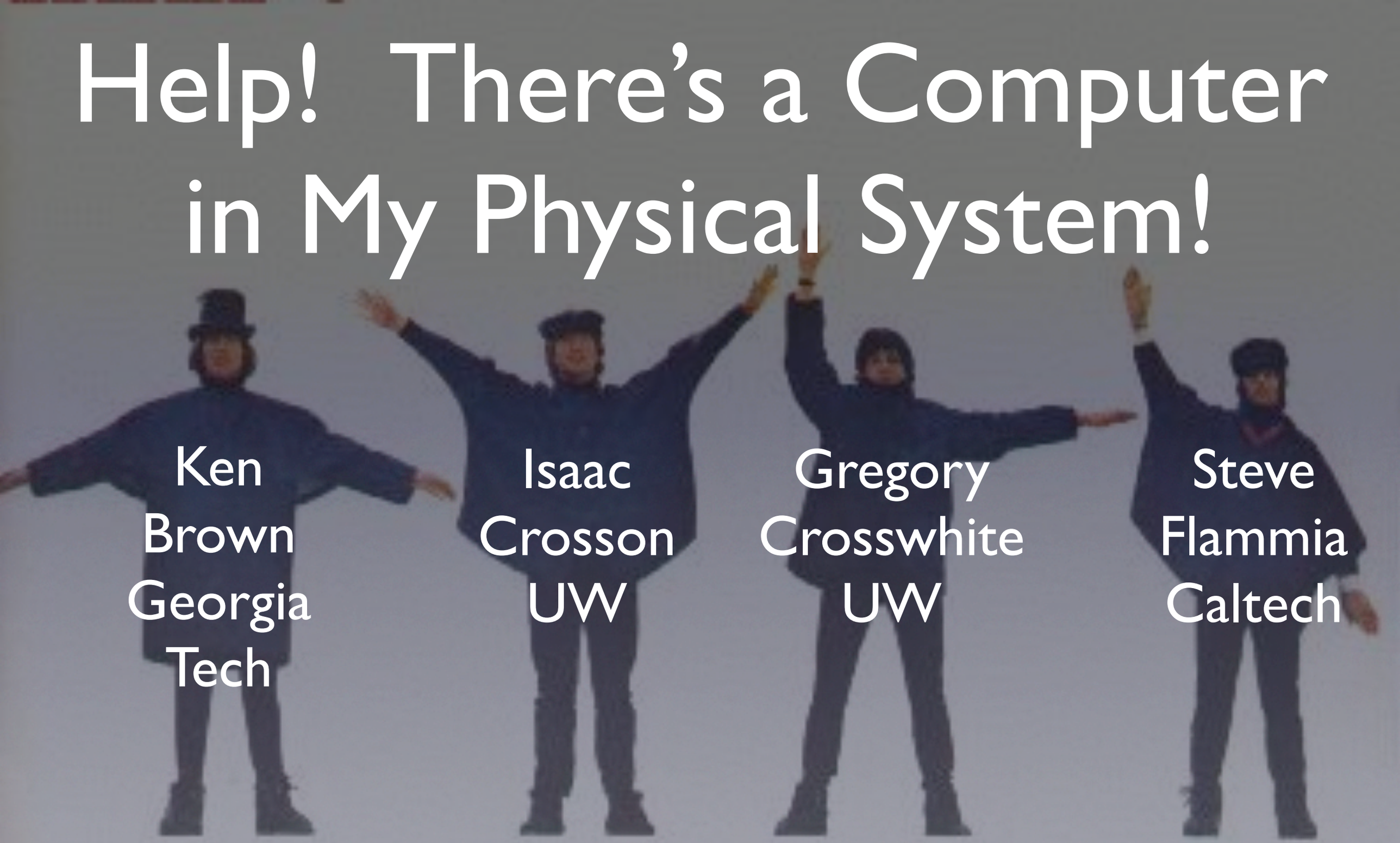
Help! There's a Computer in My Physical System!



Computation laid out spatially

Dave Bacon
University of Washington

Help! There's a Computer in My Physical System!



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